

Autonomous discovery of symmetry-protected topological phases: benchmark results

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Abstract

We benchmark a label-free autonomous agent against uniform random sampling on the task of discovering distinct topological phases of the long-range Kitaev chain in symmetry class BDI. Across a 10×4 grid of phase counts $d \in [20, 200]$ and query budgets 10^4 – 3×10^5 , the agent discovers 5–25 more phases than uniform sampling at equal budget, and the margin is *independent of budget* across a 30-fold range. The agent at 10^4 solves finds more phases than uniform sampling at 3×10^5 — a measured $30\times$ query advantage. Measuring the induced probability of each winding sector with 4×10^6 uniform draws, we find the agent reaches sectors that those draws never hit ($p < 2.5 \times 10^{-7}$): 8 such sectors at $d = 140$ from a random start, 21 from the trivial limit. We separate the rotation-invariant part of the advantage from the part that comes from knowing where the trivial limit is, and we list the claims from earlier drafts that these measurements retract.

1 Setup

After a Jordan–Wigner transformation the generalized cluster chain is the long-range Kitaev chain in class BDI,

$$H = i \sum_j \sum_{\alpha=0}^{d-1} c_{\alpha} a_j b_{j+\alpha}, \quad c = (t_0, -t_1, \dots, -t_{d-1}), \quad \|t\| = 1.$$

Class BDI is $\mathbb{Z}$ -classified in one dimension. The invariant is the winding number of $f(k) = \sum_{\alpha} c_{\alpha} e^{ik\alpha}$, equal to the number of roots of the associated polynomial inside the unit disc; it takes d values, with $e_0 \leftrightarrow \omega = 0$ (trivial paramagnet) and $e_{d-1} \leftrightarrow \omega = d - 1$. **All labels in this report are computed by exact companion-matrix root counting**, not by phase unwrapping on a k -grid, so they carry no lattice-resolution error.

In the Neveu–Schwarz sector $M(t)$ is a twisted circulant, so the ground-state covariance — the polar factor of M — is closed form,

$$g(r) = \frac{1}{L} \sum_k \frac{f(k)}{|f(k)|} e^{-ikr},$$

verified against explicit polar decomposition to 6×10^{-15} with $\|g\| = 1$ exactly. A ground state at $L = 1000$ costs 2.2 ms. Throughout: $L = 500$, lazy factor $k = 30$ (the ground state is re-solved, and one query spent, every 30th parameter step; the gradient is reused in between, and coverage is counted only at solved points).

Baseline. Uniform sampling draws t from the rotation-invariant measure (Gaussian, normalised to the unit sphere) and labels each draw. This choice matters: sampling along coordinate axes or from heavy-tailed measures presupposes the coupling basis, and the advantage measured against those baselines does not survive a hidden orthogonal rotation.

Accounting. One query = one ground-state solve, for both methods. Agent points are the median over 5 independent seeds (4 in the start-decomposition runs); uniform points are the median over 5 independent realisations. Bands are min–max, never standard errors.

2 The grid

Table 1: Median phases discovered, agent (e_0 start) / uniform, and the margin. Bold where the agent’s *lowest* seed exceeds uniform’s *highest* realisation.

d	10^4	3×10^4	10^5	3×10^5
20	19/18 (+1)	20/19 (+1)	20/20 (0)	20/20 (0)
40	32/27 (+5)	34/28 (+6)	35/30 (+5)	37/31 (+6)
60	43/32 (+11)	45/35 (+10)	47/37 (+10)	49/38 (+11)
80	53/37 (+16)	55/40 (+15)	58/43 (+15)	60/45 (+15)
100	59/42 (+17)	62/44 (+18)	67/49 (+18)	68/51 (+17)
120	64/45 (+19)	67/47 (+20)	70/52 (+18)	75/55 (+20)
140	67/47 (+20)	71/52 (+19)	78/56 (+22)	80/59 (+21)
160	74/52 (+22)	77/57 (+20)	80/59 (+21)	88/63 (+25)
180	72/54 (+18)	77/57 (+20)	86/63 (+23)	91/66 (+25)
200	78/59 (+19)	83/62 (+21)	87/67 (+20)	96/71 (+25)

In 36 of the 40 cells the agent’s lowest seed exceeds uniform’s highest realisation. The four exceptions are all $d = 20$, where both methods saturate at 20/20 from 3×10^4 upwards; at 10^4 the agent loses the worst case there (18 vs 20), because a trajectory still has to travel while 10^4 uniform draws already cover a 20-phase space.

The margin does not come from the budget. Over a 30-fold budget range the margin at fixed d is flat to within seed noise ($d = 200$: +19, +21, +20, +25). This closes the objection that the advantage is bought with compute: it is a difference in sampling efficiency, not in saturation.

3 Sampling efficiency

Table 2: The agent at 10^4 solves against uniform sampling at 3×10^5 — a $30\times$ budget difference in uniform’s favour.

d	agent @ 10^4	uniform @ 3×10^5	surplus
60	43	38	+5
100	59	51	+8
140	67	59	+8
200	78	71	+7

$30\times$ is a *measured lower bound*: 3×10^5 is the largest budget we ran. Uniform coverage grows logarithmically in budget (four-point fit, $R^2 = 0.95\text{--}0.99$, slope 0.6–3.7 phases per e -fold); extrapolating that fit, uniform would need $10^6\text{--}10^7$ draws to match the agent at 10^4 . The extrapolation assumes the logarithmic law survives past 3×10^5 and ignores the hard ceiling at d ; only the $30\times$ is measured, and only the $30\times$ is claimed.

4 Where the margin comes from

An earlier draft concluded that “the advantage is the corner start, not the exploration”, generalising from $d = 60$ alone. That generalisation was wrong.

At $d = 200$ the basis-free component exceeds the e_0 bonus. Worst-case separation for the random start holds at $d \geq 100$ ($d = 200$: worst agent seed 73 > best uniform realisation 71; $d = 140$: 60 > 58; $d = 100$: 51 > 49); $d = 60$ is a boundary tie (38 = 38) — exactly the cell the earlier draft generalised from.

Label-free phase discovery beats uniform sampling by $\sim 30 \times$ in queries, and the gap widens with d

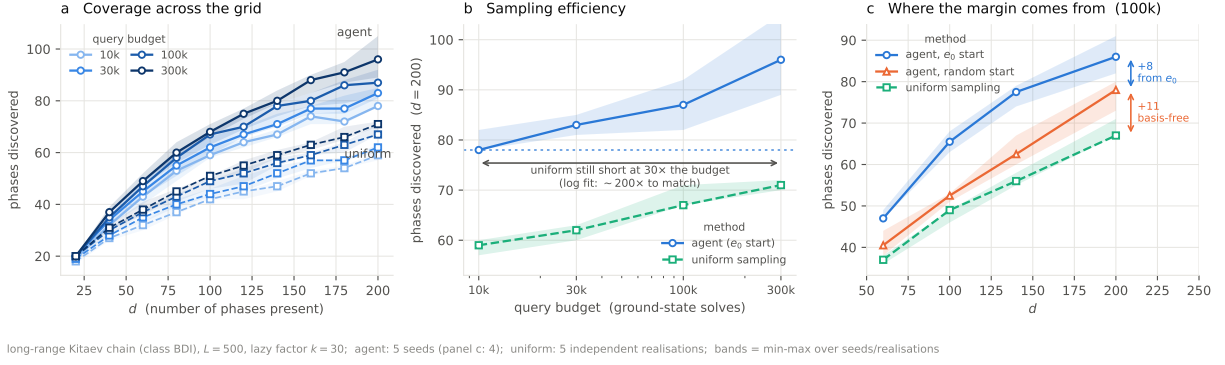


Figure 1: (a) Coverage across the grid; agent solid, uniform dashed, colour encodes budget. (b) Coverage against budget at $d = 200$: the agent’s 10^4 point lies above uniform’s 3×10^5 point. (c) Decomposition of the margin by starting point.

Table 3: Start decomposition at 10^5 solves, median over 4 seeds. The basis-free component is the margin a *random* start achieves; it grows with d and is not capped.

d	agent, e_0	agent, random	uniform	basis-free	extra from e_0
60	47.0	40.5	37	+3.5	+6.5
100	65.5	52.5	49	+3.5	+13.0
140	77.5	62.5	56	+6.5	+15.0
200	86.0	78.0	67	+11.0	+8.0

The margin is therefore a sum of two terms: a rotation-invariant exploration gain that grows with d , and an additional gain from starting at the trivial limit. The second is legitimate physics — any experiment starts from the decoupled limit — but it is basis-dependent knowledge and is labelled as such throughout.

The two starts are redundant but not identical. At $d = 200$ the e_0 union over 4 seeds is 103 sectors, the random-start union 93, their intersection 86 and their union 110 (Jaccard 0.78): e_0 extends into the low tail, random starts into the high tail.

5 The agent reaches rare sectors

Uniform sampling induces a sharply peaked measure p_ω on the d winding sectors. We measured it with 4×10^6 draws per d , giving a detection floor of 2.5×10^{-7} .

Table 4: Sectors reached. “never seen” counts sectors the agent reached that 4×10^6 uniform draws never hit ($p < 2.5 \times 10^{-7}$); $\omega = 0$ is excluded from the e_0 column, since it is the starting point rather than a discovery.

d	uniform		agent, e_0 @ 10^5		agent, random @ 10^5	
	@ 10^5	@ 4×10^6	reached	never seen	reached	never seen
60	37	43	51	7	45	3
100	49	56	74	18	58	5
140	56	66	88	21	74	8

Two checks. *Are the rare labels artefacts?* No. Labels come from exact root counting, and the structure is coherent: uniform’s support is a near-contiguous band and every extra sector the

The agent reaches sectors that four million uniform draws never hit

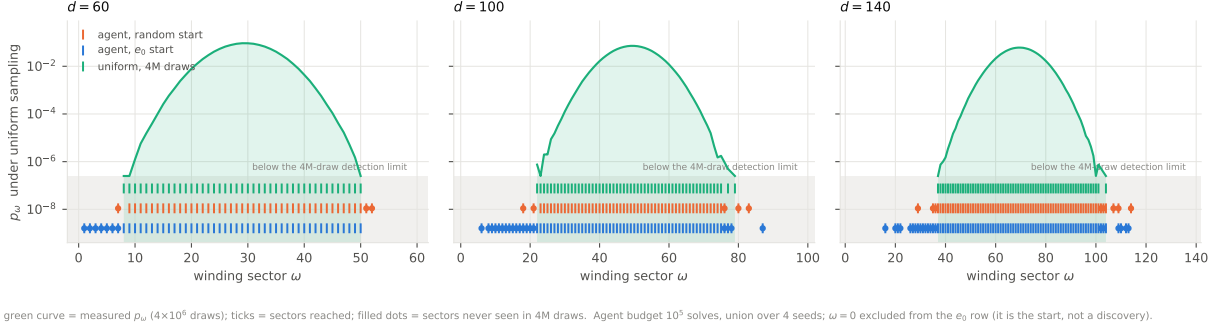


Figure 2: The measured sector distribution and the reach of each method. Filled dots mark sectors never seen in 4×10^6 uniform draws.

agent reaches sits on that band’s shoulders. Label noise would produce scattered outliers, not a clean extension. *Is each rare find a single lucky seed?* For the e_0 start, no: 14 of the 21 extra sectors at $d = 140$ were found independently by at least 2 of 4 seeds (7 of 7 at $d = 60$). For the random start the individual finds are lottery-like (2 of 8 repeated), though the fact that *some* tail sector is reached is stable.

How rare. The measured statement is $p < 2.5 \times 10^{-7}$. A quadratic fit to $\log p_\omega$ (residual rms 0.03–0.04 in log space over the measured range) extrapolates to far smaller values in the extreme tail, but that is an extrapolation over ten orders of magnitude with no reason for the large-deviation rate function to remain Gaussian. It is not claimed here.

Mechanism, stated plainly. The agent does not snipe isolated exotic sectors. Both methods cover a contiguous band around the mode; the agent’s band is wider, and the sectors at its edge are quantitatively rare. Both halves of that sentence are needed.

6 Limitations and retractions

- **Retracted:** “the advantage is the corner start, not the exploration.” Superseded by the start decomposition above.
- **Retracted:** the earlier crossover result. That arm used a memory built on the accumulated phase ϕ , which is topologically complete ($\phi_{\text{total}} = 2\pi\omega$) and therefore label-equivalent.
- **Not isolated:** “an underfit autoencoder explores better.” L , bootstrap count, latent dimension and autoencoder epochs have each been ruled out as the cause of the observed difference; hidden width 32 vs 64 remains untested. Do not cite this claim.
- **Scope:** everything here exploits exact solvability of the free-fermion chain. The interacting (κ) case needs differentiable DMRG, which is established in the literature but not done here.
- **Scope:** $d = 20$ carries no signal — both methods saturate.
- **Open:** $d = 200$ sector statistics were still running when this was compiled; the rarity table covers $d \in \{60, 100, 140\}$.

7 Reproduction

<code>final_res/f*.json</code>	agent runs, all d and budgets
<code>unif_res/u*.json, u1*.json</code>	uniform baselines (consolidated in <code>unif_all.json</code>)
<code>which_res/w*.json</code>	start-type runs, with full sector sets
<code>hist_res/h*.json</code>	sector histograms, 4×10^6 draws per d
<code>plot_main.py, plot_rarity.py</code>	the two figures
<code>REPORT.md §23</code>	full narrative, including superseded sections
